

# Introduction to GLMs

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## 1 Introduction



### Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
```

```
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

## 2 Welcome

Welcome to Epidemiology 204: Quantitative Epidemiology III (Statistical Models).

Epi 204 is a course on **regression modeling**.

## 3 What you should already know

### Warning

Epi 202, Epi 203, and Sta 108 are prerequisites for this course. If you haven't passed one of these courses, talk to me ASAP.

### 3.0.1 Epi 202: probability models

- Probability distributions
  - binomial
  - Poisson

- Gaussian
- exponential

- 
- Characteristics of probability distributions
    - Mean, median, mode, quantiles
    - Variance, standard deviation, overdispersion
- 

- Characteristics of samples
  - independence, dependence, covariance, correlation
  - ranks, order statistics
  - identical vs nonidentical distribution (homogeneity vs heterogeneity)
  - Laws of Large Numbers
  - Central Limit Theorem for the mean of an iid sample

### 3.0.2 Epi 203: inference for one or several homogenous populations

- the maximum likelihood inference framework:
    - likelihood functions
    - log-likelihood functions
    - score functions
    - estimating equations
    - information matrices
    - point estimates
    - standard errors
    - confidence intervals
    - hypothesis tests
    - p-values
- 
- Hypothesis tests for one, two, and  $>2$  groups:
    - t-tests/ANOVA for Gaussian models
    - chi-square tests for binomial and Poisson models
    - nonparametric tests:
      - \* Wilcoxon signed-rank test for matched pairs
      - \* Mann–Whitney/Kruskal-Wallis rank sum test for  $\geq 2$  independent samples
      - \* Fisher’s exact test for contingency tables
      - \* Cochran–Mantel–Haenszel–Cox log-rank test
- 

For all of the quantities above, and especially for confidence intervals and p-values, you should know how **both**:

- how to compute them
  - how to interpret them
- 

### 3.0.3 Stat 108: linear regression models

- building models for Gaussian outcomes
  - multiple predictors
  - interactions
- regression diagnostics
- fundamentals of R programming; e.g.:
  - Wickham et al. (2023)
  - Dalgaard (2008)

- RMarkdown or Quarto for formatting homework<sup>1</sup>
  - LaTeX for writing math in RMarkdown/Quarto

## 4 What we will cover in this course

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- Linear (Gaussian) regression models (review and more details)
- Regression models for non-Gaussian outcomes
  - binary
  - count
  - time to event
- Statistical analysis using R

We will start where Epi 203 left off: with linear regression models.

## 5 Motivations for regression models

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**Exercise 5.1.** Why do we need regression models?

---

*Solution 5.1.*

- when there's not enough data to analyze every subgroup of interest individually
- especially when subgroups are defined using continuous predictors

### 5.1 Example: Adelie penguins

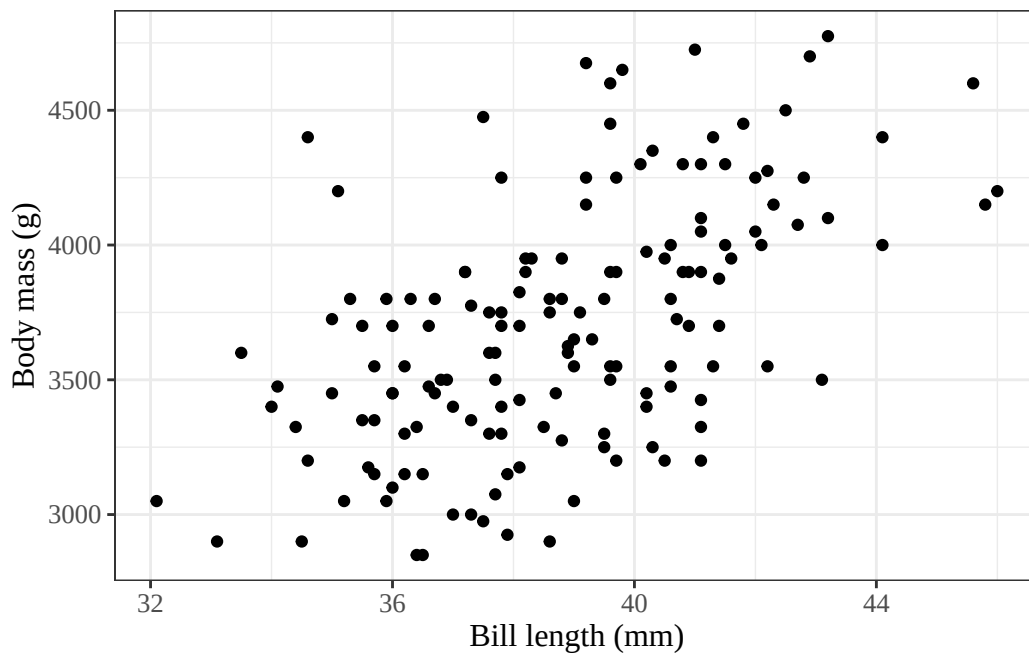


Figure 1: Palmer penguins

---

<sup>1</sup><https://r4ds.hadley.nz/quarto>

## 5.2 Linear regression

```
ggpenguins2 <-  
  ggplot(ggpenguins) +  
    stat_smooth(  
      method = "lm",  
      formula = y ~ x,  
      geom = "smooth"  
    )  
  
ggpenguins2 |> print()
```

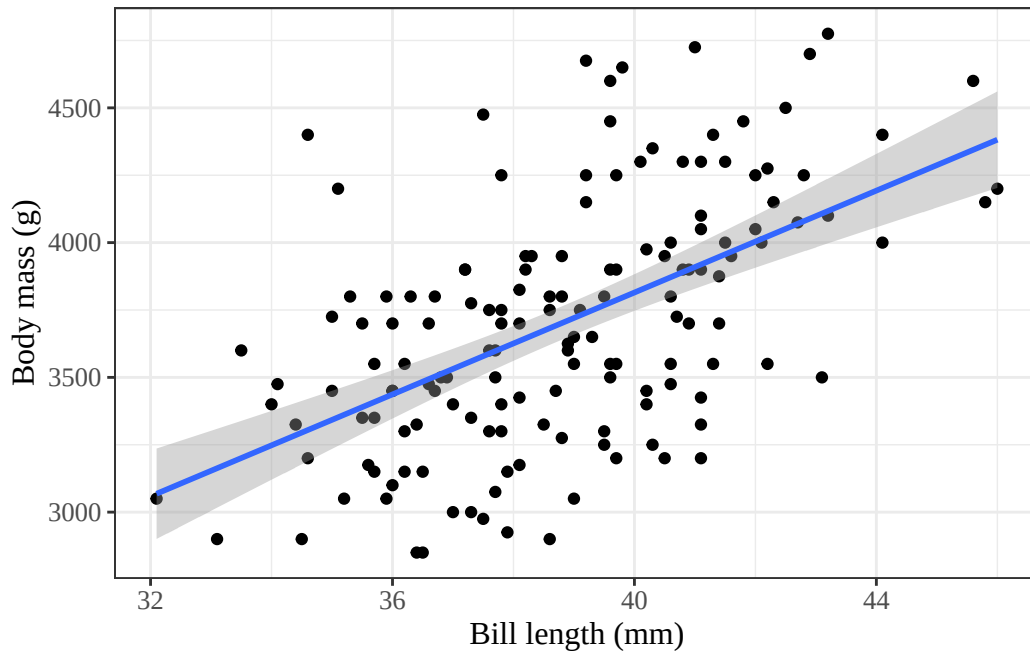


Figure 2: Palmer penguins with linear regression fit

## 5.3 Curved regression lines

```
ggpenguins2 <- ggplot(ggpenguins) +  
  stat_smooth(  
    method = "lm",  
    formula = y ~ log(x),  
    geom = "smooth"  
  ) +  
  xlab("Bill length (mm)") +  
  ylab("Body mass (g)")  
ggpenguins2
```

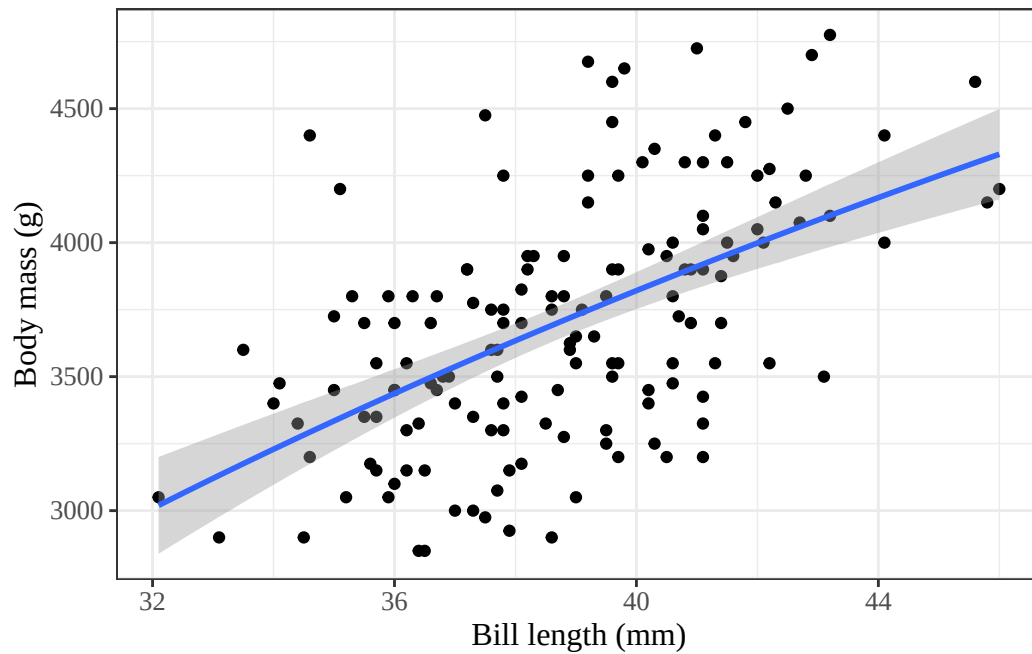


Figure 3: Palmer penguins - curved regression lines

## 5.4 Multiple regression

```
ggpenguins <-
  palmerpenguins::penguins |>
  ggplot(
    aes(
      x = bill_length_mm,
      y = body_mass_g,
      color = species
    )
  ) +
  geom_point() +
  stat_smooth(
    method = "lm",
    formula = y ~ x,
    geom = "smooth"
  ) +
  xlab("Bill length (mm)") +
  ylab("Body mass (g)")
ggpenguins |> print()
```



Figure 4: Palmer penguins - multiple groups

## 5.5 Modeling non-Gaussian outcomes

```
library(glmx)
data(BeetleMortality)
beetles <- BeetleMortality |>
  mutate(
    pct = died / n,
    survived = n - died
  )

plot1 <-
  beetles |>
  ggplot(aes(x = dose, y = pct)) +
  geom_point(aes(size = n)) +
  xlab("Dose (log mg/L)") +
  ylab("Mortality rate (%)") +
  scale_y_continuous(labels = scales::percent) +
  # xlab(bquote(log[10]), bquote(CS[2])) +
  scale_size(range = c(1, 2))

print(plot1)
```

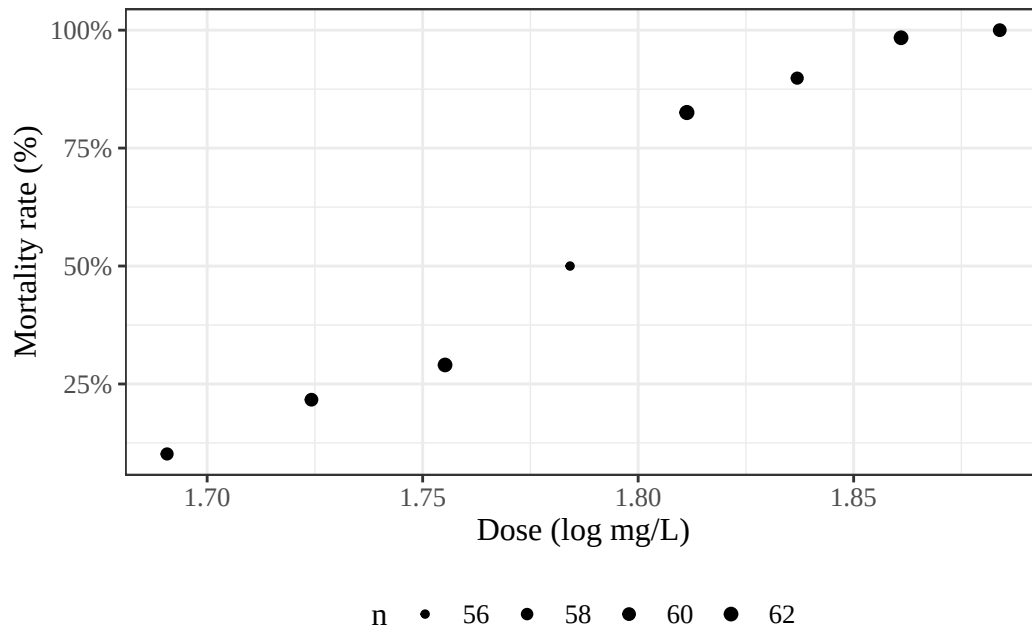


Figure 5: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)



## 5.6 Why don't we use linear regression?

```
beetles_long <-  
  beetles |>  
  reframe(  
    .by = everything(),  
    outcome = c(  
      rep(1, times = died),  
      rep(0, times = survived)  
    )  
  )  
  
lm1 <-  
  beetles_long |>  
  lm(  
    formula = outcome ~ dose,  
    data = _  
  )  
  
range1 <- range(beetles$dose) + c(-.2, .2)  
  
f_linear <- function(x) predict(lm1, newdata = data.frame(dose = x))  
  
plot2 <-  
  plot1 +  
  geom_function(fun = f_linear, aes(col = "Straight line")) +  
  labs(colour = "Model", size = "")  
print(plot2)
```



Figure 6: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

## 5.7 Zoom out

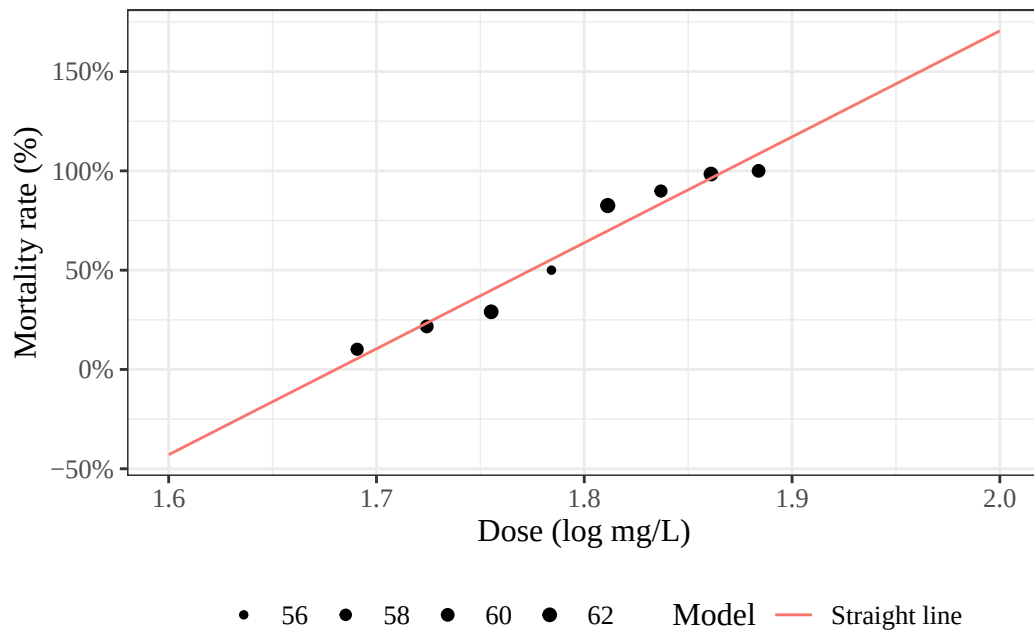


Figure 7: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

## 5.8 log transformation of dose?

```
lm2 <-  
  beetles_long |>  
  lm(formula = outcome ~ log(dose), data = _)  
  
f_linearlog <- function(x) predict(lm2, newdata = data.frame(dose = x))  
  
plot3 <- plot2 +  
  expand_limits(x = c(1.6, 2)) +  
  geom_function(fun = f_linearlog, aes(col = "Log-transform dose"))  
  
print(plot3 + expand_limits(x = c(1.6, 2)))
```



Figure 8: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

## 5.9 Logistic regression

```
glm1 <- beetles |>
  glm(formula = cbind(died, survived) ~ dose, family = "binomial")

f <- function(x) {
  glm1 |>
    predict(newdata = data.frame(dose = x), type = "response")
}

plot4 <- plot3 + geom_function(fun = f, aes(col = "Logistic regression"))
print(plot4)
```



Figure 9: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

## 6 Structure of regression models

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**Exercise 6.1.** What is a regression model?

---

**Definition 6.1** (Regression model). Regression models are conditional probability distribution models:

$$P(Y|\tilde{X})$$

---

**Exercise 6.2.** What are some of the names used for the variables in a regression model  $P(Y|\tilde{X})$ ?

---

**Definition 6.2** (Outcome). The outcome variable in a regression model is the variable whose distribution is being described; in other words, the variable on the left-hand side of the “|” (“pipe”) symbol.

The outcome variable is also called the **response variable**, **regressand**, **predicted variable**, **explained variable**, **experimental variable**, **output variable**, **dependent variable**, **endogenous variables**, **target**, or **label**.

and is typically denoted  $Y$ .

---

**Definition 6.3** (Predictors). The predictor variables in a regression model are the conditioning variables defining subpopulations among which the outcome distribution might vary.

Predictors are also called **regressors**, **covariates**, **independent variables**, **explanatory variables**, **risk factors**, **exposure variables**, **input variables**, **exogenous variables**, **candidate variables** (Dunn and Smyth (2018)), **carriers** (Dunn and Smyth (2018)), **manipulated variables**, or **features** and are typically denoted  $\tilde{X}$ .<sup>2</sup>

---

Table 1: Common pairings of terms for variables  $\tilde{X}$  and  $Y$  in regression models  $P(Y|\tilde{X})$ <sup>4</sup>

| $\tilde{X}$ | $Y$                   | usual context                     |
|-------------|-----------------------|-----------------------------------|
| input       | output                |                                   |
| independent | dependent             |                                   |
| predictor   | predicted or response |                                   |
| explanatory | explained             |                                   |
| exogenous   | endogenous            | econometrics                      |
| manipulated | measured              | randomized controlled experiments |
| exposure    | outcome               | epidemiology                      |
| feature     | label or target       | machine learning                  |

---



---

**Exercise 6.3.** What is the general structure of a generalized linear model?

---

*Solution 6.1.* Generalized linear models have three components:

1. The **outcome distribution** family:  $p(Y|\mu(\tilde{x}))$
  2. The **link function**:  $g(\mu(\tilde{x})) = \eta(\tilde{x})$
  3. The **linear component**:  $\eta(\tilde{x}) = \tilde{x} \cdot \beta$
- 

1. The **outcome distribution** family (a.k.a. the **random component** of the model)

- Gaussian (normal)
  - Binomial
  - Poisson
  - Exponential
  - Gamma
  - Negative binomial
- 

2. The **linear component** (a.k.a. the *linear predictor* or *linear functional form*) describing how the covariates combine to define subpopulations:

$$\eta(\tilde{x}) \stackrel{\text{def}}{=} \tilde{x}^\top \tilde{\beta} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots$$

---

<sup>2</sup>The “~” (“tilde”) symbol in the notation  $\tilde{X}$  indicates that  $\tilde{X}$  is a vector. See the appendices<sup>3</sup> for a table of notation used in these notes.

<sup>4</sup>adapted from [https://en.wikipedia.org/wiki/Dependent\\_and\\_independent\\_variables#Synonyms](https://en.wikipedia.org/wiki/Dependent_and_independent_variables#Synonyms)

---

3. The **link function** relating the outcome distribution to the linear component, typically through the mean:

- identity:  $\mu(y) = \eta(\tilde{x})$
- logit:  $\log\left\{\frac{\mu(y)}{1-\mu(y)}\right\} = \eta(\tilde{x})$
- log:  $\log\{\mu(y)\} = \eta(\tilde{x})$
- inverse:  $(\mu(y))^{-1} = \eta(\tilde{x})$
- clog-log:  $\log\{-\log\{1 - \mu(y)\}\} = \eta(\tilde{x})$

Components 2 and 3 together are sometimes called the **systematic component** of the model (for example, in Dunn and Smyth (2018)).

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Dalgaard, Peter. 2008. *Introductory Statistics with r*. New York, NY: Springer New York. <https://link.springer.com/book/10.1007/978-0-387-79054-1>.

Dunn, Peter K, and Gordon K Smyth. 2018. *Generalized Linear Models with Examples in R*. Vol. 53. Springer. <https://link.springer.com/book/10.1007/978-1-4419-0118-7>.

Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science*. "O'Reilly Media, Inc.". <https://r4ds.hadley.nz/>.